

Lecture 11: Implementation

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English version of slides by ©Bo Wahlberg



Today's Ordlista

English	Svenska
steam engine, windmill	ångmaskin, väderkvarn
converter	omvandlare
Real-time system	Realtidssystem
kontinuerliga signalen	continuous signal
discrete time points	diskreta tidpunkter
sampling interval	samplingintervall
piecewise constant	styckvis konstant
sample and hold	hållkrets
user interface	användargränssnitt
synchronization	synkronisering
autocode generation	autocodegenerering
difference approximations	differensapproximationer
digital implementation	digital implementering
backward Euler, forward Euler	Euler bakåt, Euler framåt
the differentiation operator	deriveringsoperatör
the shift operator	förskjutningsoperatör
the inverse shift operator	inverterade förskjutningsoperatör
sampling interval	samlingsintervall
The aliasing effect (folding distortion)	Aliaseffekten (vikningsdistorion)



Outline

Recap

Analog implementation

Digital implementation

Real-time system

Control algorithm

Difference approximations

Control algorithm: Examples

Sampling interval

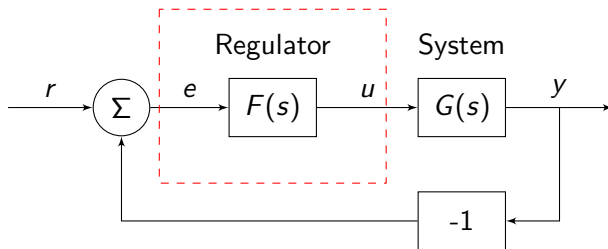
Recap

In the previous lectures (F7, F8, F9)..

- State-space model (with info: (A, B, C, D) , $x(0)$)
- Observability and Controllability (observable and controllable states)
- State feedback (from $x(t)$) and state feedback with observer (from $y(t)$)

Today:

- Implementation:



How does one realize the connection between $e(t)$ and $u(t)$?



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Analog implementation

There are different types of implementations:

- Mechanical
- Hydraulic
- Pneumatic
- Electronic

The **centrifugal governor**

(Centrifugalregulatorn) is a mechanical implementation of a P-controller

- James Watt (1788) for the steam engine
- Christiaan Huygens over 100 years earlier for the windmill

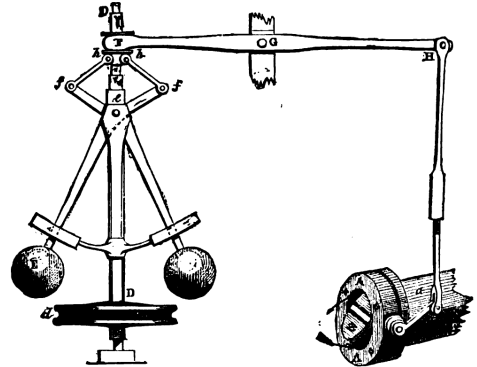


FIG. 4.—Governor and Throttle-Valve.

Source: [wiki:Centrifugal governor](https://en.wikipedia.org/wiki/Centrifugal_governor)

Analog implementation: the centrifugal governor



Sources: (left), (right top), (right bottom left) (right bottom right)



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1. **A/D:** Measures the continuous signal at discrete time points (the distance between these is called the *sampling interval* and is denoted by T : $T = t_{k+1} - t_k$).
2. Computer program = "control algorithm"

3. **D/A:** Sends out the computed (discrete) control signal. Most often, it is held piecewise constant between the data points (the simplest form of interpolation). This is called a sample and hold circuit.

8



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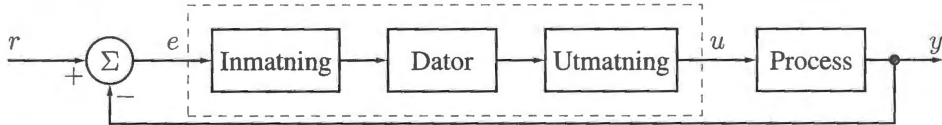
Difference approximations

Control algorithm: Examples

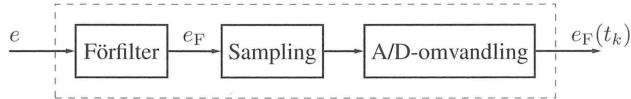
Sampling interval

Real-time system

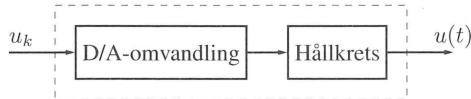
Typical computer implementation [Fig. 11.1]:



Input/Inmatning to the computer [Fig. 11.2]

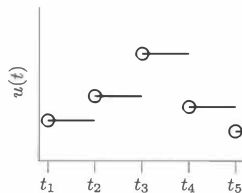
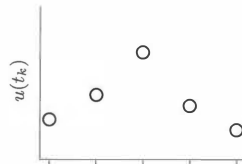
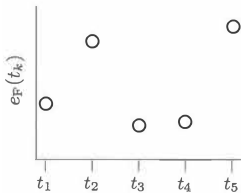
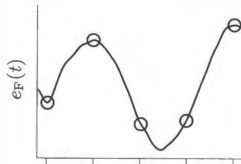
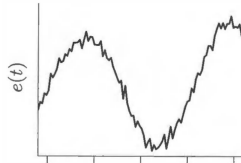


Output/Utmatning from the computer [Fig. 11.3]



Real-time system

Signals in Figs. 11.1–11.3 [Fig. 11.4]



Real-time system

- ▶ User interface
- ▶ Synchronization
- ▶ Communication (latency, wireless, 5G, ..)

Autocode generation (Matlab $\rightarrow$ C)

Compare the water tanks!



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- Difference approximations

- Control algorithm: Examples

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► **PI:** $u = K\left(e + \frac{1}{T_I} \int e\right) \Rightarrow \dot{u} = K\dot{e} + \frac{K}{T_I}e$

Digital implementation is done through the *numerical* solution of differential equations.

The idea is to replace derivatives with (finite) difference approximations.

Difference operator formalism

- The differentiation operator, p .

$$pu(t) = \dot{u}(t)$$

- The shift operator, q_T .

$$q_T u(t) = u(t + T)$$

- The inverse shift operator, q_T^{-1} .

$$q_T^{-1} u(t) = u(t - T)$$

Backward Euler

We can approximate the first derivative with

$$\dot{u}(t) \approx \Delta_E u(t) = \frac{u(t) - u(t - T)}{T}$$

Using operators, this can be written as

$$p \approx \frac{1}{T} (1 - q_T^{-1})$$

This is *backward Euler*.

Homework: Try to prove why $\ddot{u}(t) \approx \frac{1}{T^2} (u(t) - 2u(t - T) + u(t - 2T))$

$$p \approx \frac{2}{T} \frac{1 - q_T^{-1}}{1 + q_T^{-1}}$$

► Method 1: `jmf [(A.7)]`

$$q_T^{-1} = e^{-pT} = \frac{e^{-pT/2}}{e^{pT/2}} \approx \frac{1 - pT/2}{1 + pT/2}$$

$$y(t) = \int_0^t u(\tau) d\tau = y(t-T) + \int_{t-T}^t u(\tau) d\tau$$
$$y(t) = y(t - T) + T \frac{u(t) + u(t - T)}{2} \Rightarrow \frac{1}{p} \approx \frac{T}{2} \frac{1 + q_T^{-1}}{1 - q_T^{-1}}$$



Corresponds exactly to the integration of piecewise constant interpolation

Exact expression for a piecewise-constant input signal

Discrete-time system

18

Expressing the shift operator as $q_T = e^{pT}$

Let us consider a Taylor series expansion

$$q_T u(t) = u(t + T) = \{\text{Taylor}\} = u(t) + Tpu(t) + \dots = e^{pT} u(t)$$

We see that we can express the shift operator as

$$q_T = e^{pT} \quad q_T^{-1} = e^{-pT}$$

which can be compared with the time shift e^{sT} .

- ▶ Euler backwards: $\frac{1}{T} \left(1 - e^{-pT} \right) = p + \mathcal{O}(T)$
- ▶ Tustin's formula: $\frac{2}{T} \left(\frac{1 - e^{-pT}}{1 + e^{-pT}} \right) \approx p + \mathcal{O}(T^2)$



We can rewrite a given differential equation

using backward Euler as

from which we can then solve for $u(t)$ as

$$u(t) = \frac{\beta\tau_D}{\beta\tau_D + T} \underbrace{\left(q_T^{-1} u(t) \right)}_{u(t-T)} + \frac{\tau_D + T}{\beta\tau_D + T} e(t) - \frac{\tau_D}{\beta\tau_D + T} \underbrace{q_T^{-1} e(t)}_{e(t-T)}$$

Control algorithm: Examples

Example (PI + Tustin)

Given a differential equation $\dot{u} = K \left(\dot{e} + \frac{1}{T_I} e \right)$

We identify from the relationship $U = FE$ that: $F(p) = \frac{K \left(p + \frac{1}{T_I} \right)}{p}$, which we can rewrite using Tustin's formula as

$$F_T(q_T^{-1}) = \frac{K \left(\frac{2}{T} \left(\frac{1 - q_T^{-1}}{1 + q_T^{-1}} \right) + \frac{1}{T_I} \right)}{\frac{2}{T} \left(\frac{1 - q_T^{-1}}{1 + q_T^{-1}} \right)} = \frac{K \left(1 + \frac{T}{2T_I} + \left(-1 + \frac{T}{2T_I} \right) q_T^{-1} \right)}{1 - q_T^{-1}}$$

We then find $u(t)$ from the relation: $u(t) = F_T(q_T^{-1})e(t)$

Hence, with the inserted expression for $F_T(q_T^{-1})$, we obtain

$$u(t) = u(t - T) + K \left(\left(1 + \frac{T}{2T_I} \right) e(t) + \left(-1 + \frac{T}{2T_I} \right) e(t - T) \right)$$



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How should the sampling interval be chosen?

One should take into account:

- ▶ Computational capacity
- ▶ Stability margins
- ▶ Aliasing effects

Stability margin:

Degrades in digital implementation

- ▶ Time delay due to sample-and-hold
- ▶ Approximations

Sample-and-hold $\approx T$ time delay

Example

If we want to tolerate a maximum 20° reduction in φ_m :

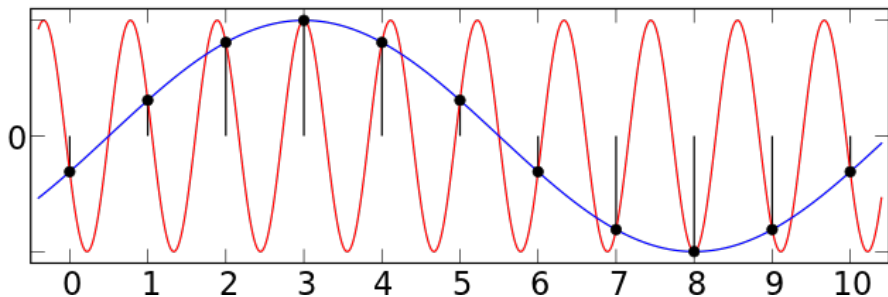
$$\implies \omega_c T < 0.35 \text{ (rad)} \implies T < \frac{0.35}{\omega_c}$$

This gives a *sampling frequency* of

$$\omega_s = \frac{2\pi}{T} \approx 20 \omega_c$$

Using a smarter discrete design, one can choose $\omega_s \approx (5 \rightarrow 10) \omega_c$

The aliasing effect (folding distortion)



After sampling, signals (red) with a frequency higher than $\frac{\omega_s}{2}$ (= the **Nyquist frequency**) **cannot** be distinguished from signals (blue) with a frequency lower than $\frac{\omega_s}{2}$.

The aliasing effect (folding distortion)

Countermeasures:

1. Determine the interval $[0, \omega_{\max}]$ where an accurate representation of the signals is needed
2. Choose T so that $\omega_{\max} < \frac{\omega_s}{2} = \text{Nyquist frequency} \implies T < \frac{\pi}{\omega_{\max}}$
3. Low-pass filter (LP) the signals *before* sampling